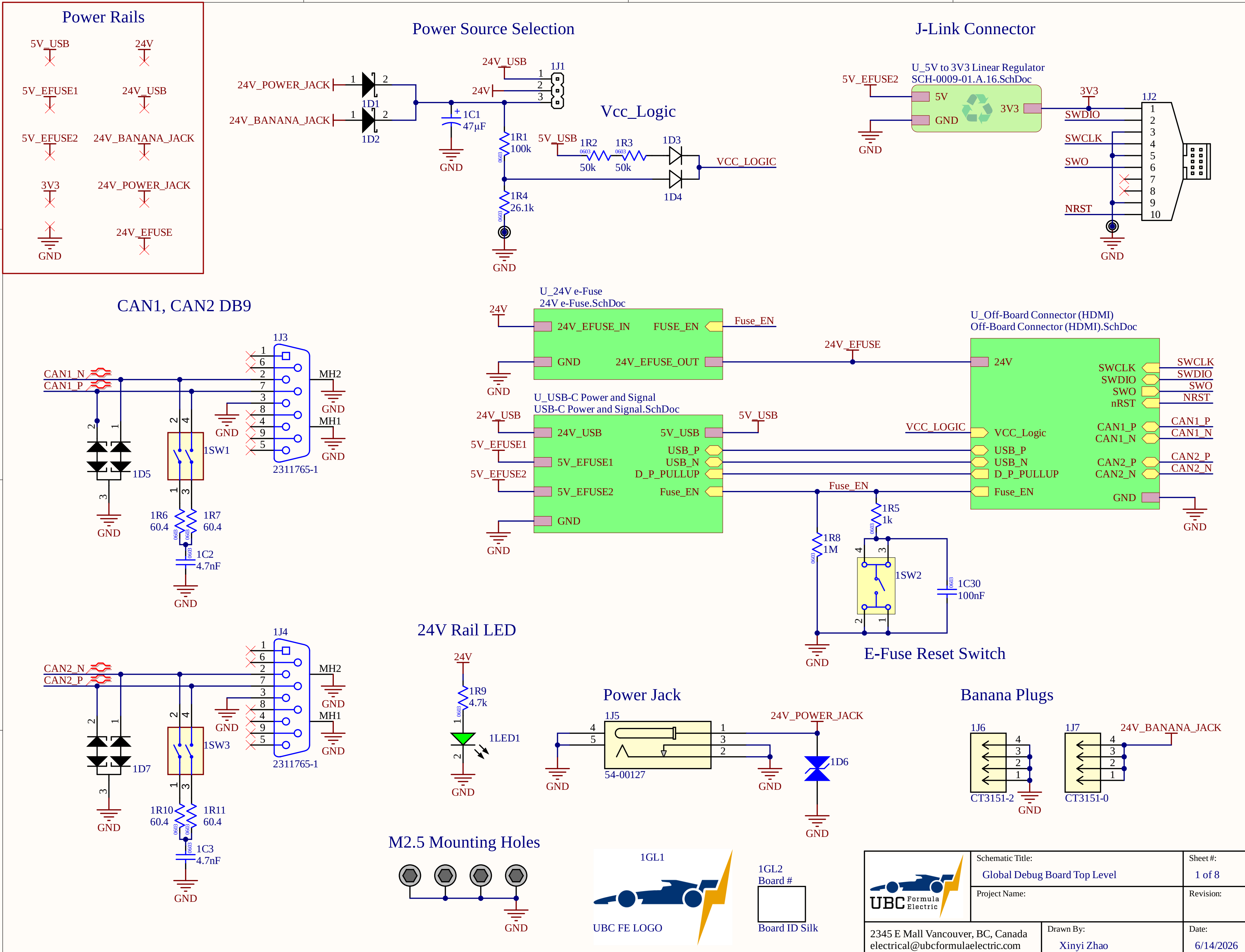
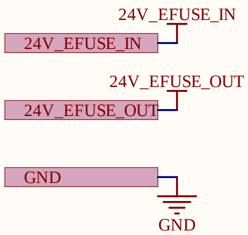
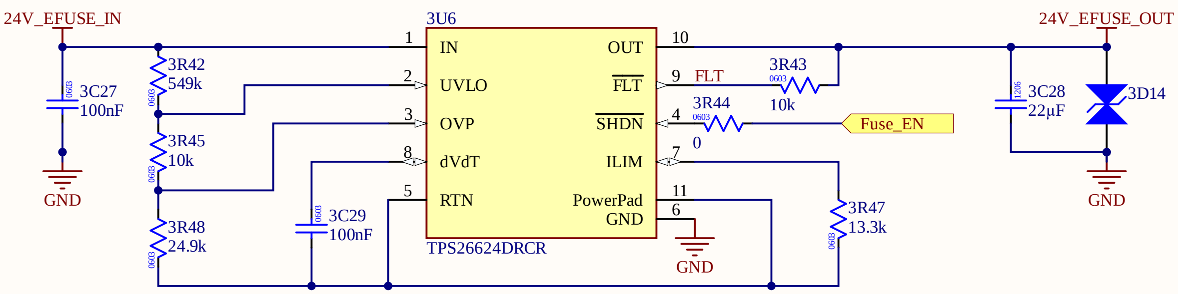

Power Rails
5V_USB
24V
5V_EFUSE1
24V_USB
5V_EFUSE2
24V_BANANA_JACK
3V3
24V_POWER_JACK
GND
24V_EFUSE
Power Source Selection
24V_POWER_JACK
24V_BANANA_JACK
1D1
1D2
24V_USB
24V
1J1
1C1 47µF
1R1 100k
1R4 26.1k
Vcc_Logic
5V_USB
1R2 50k
1R3 50k
1D3
1D4
VCC_LOGIC
J-Link Connector
U_5V to 3V3 Linear Regulator
SCH-0009-01.A.16.SchDoc
5V_EFUSE2
5V
GND
3V3
1J2
SWDIO
SWCLK
SWO
NRST
CAN1, CAN2 DB9
CAN1_N
CAN1_P
1J3
MH2
MH1
1D5
1SW1
1R6 60.4
1R7 60.4
1C2 4.7nF
2311765-1
U_24V e-Fuse
24V e-Fuse.SchDoc
24V_EFUSE_IN
FUSE_EN
Fuse_EN
24V_EFUSE_OUT
24V_EFUSE
U_Off-Board Connector (HDMI)
Off-Board Connector (HDMI).SchDoc
24V
SWCLK
SWDIO
SWO
nRST
VCC_Logic
CAN1_P
CAN1_N
CAN2_P
CAN2_N
USB_P
USB_N
D_P_PULLUP
Fuse_EN
U_USB-C Power and Signal
USB-C Power and Signal.SchDoc
24V_USB
5V_USB
5V_EFUSE1
5V_EFUSE2
1R5 1k
1R8 1M
1SW2
1C30 100nF
E-Fuse Reset Switch
CAN2_N
CAN2_P
1J4
1D7
1SW3
1R10 60.4
1R11 60.4
1C3 4.7nF
24V Rail LED
1R9 4.7k
1LED1
Power Jack
1J5
54-00127
24V_POWER_JACK
1D6
Banana Plugs
1J6
CT3151-2
1J7
CT3151-0
24V_BANANA_JACK
M2.5 Mounting Holes
1GL1
UBC FE LOGO
1GL2
Board #
Board ID Silk
UBC Formula Electric
Schematic Title: Global Debug Board Top Level
Sheet #: 1 of 8
Project Name:
Revision:
2345 E Mall Vancouver, BC, Canada
electrical@ubcformulaelectric.com
Drawn By: Xinyi Zhao
Date: 6/14/2026

Power Rails

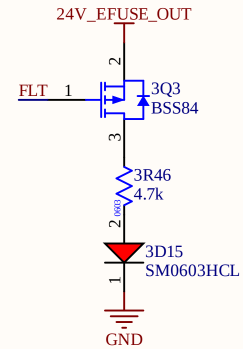


Vuvlo ~ 20V
Vovp ~ 28V
Ilim = 0.5A

FLT is active-low to indicate any faults (similar to PG).

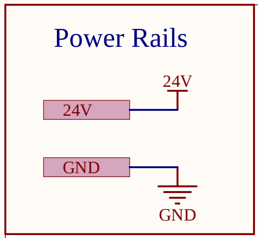


Fault Indicator LED



The dVdt pin will be DNP capacitors and have 0603 capacitor pads for testing.

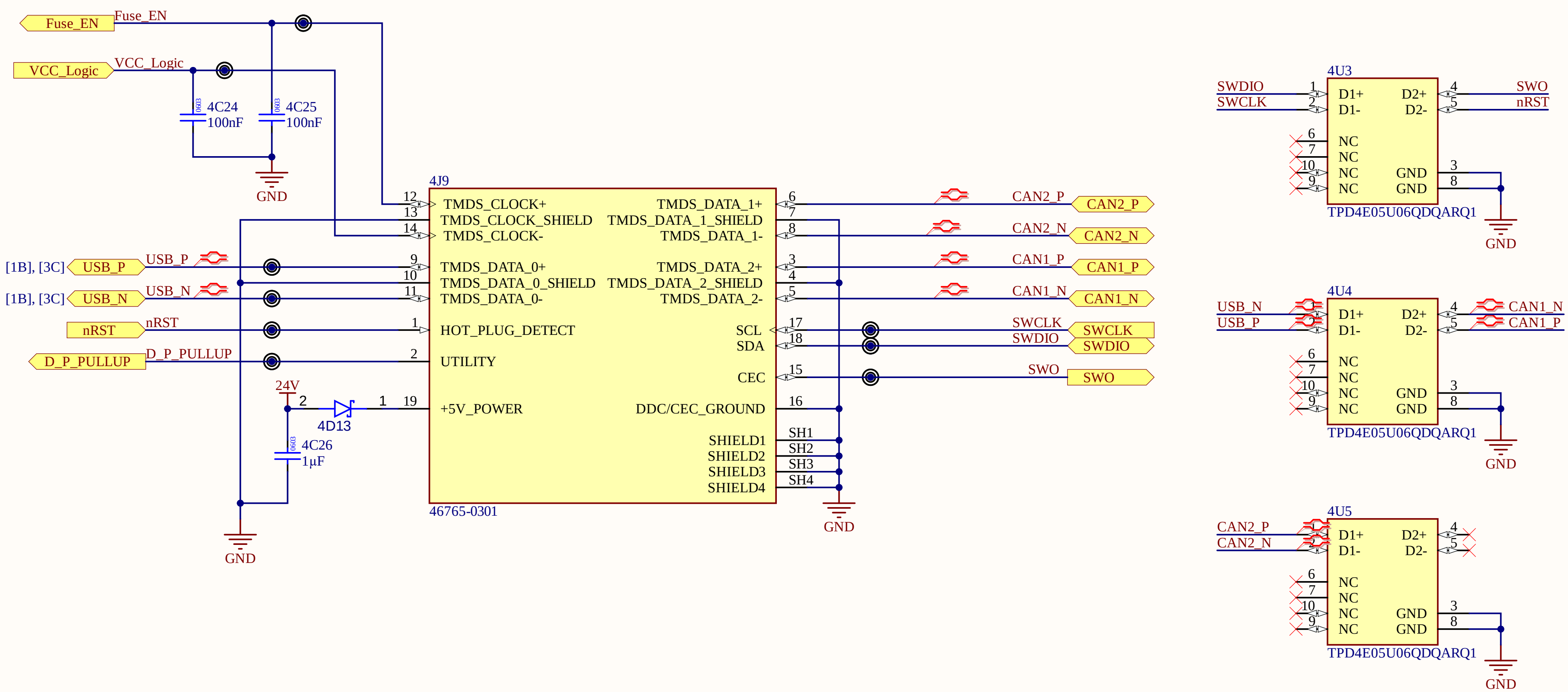
UBC Formula Electric	Schematic Title: 24V/0.5A e-Fuse	Sheet #: 7 of 8
	Project Name:	Revision:
2345 E Mall Vancouver, BC, Canada electrical@ubcformulaelectric.com	Drawn By: Xinyi Zhao	Date: 6/14/2026



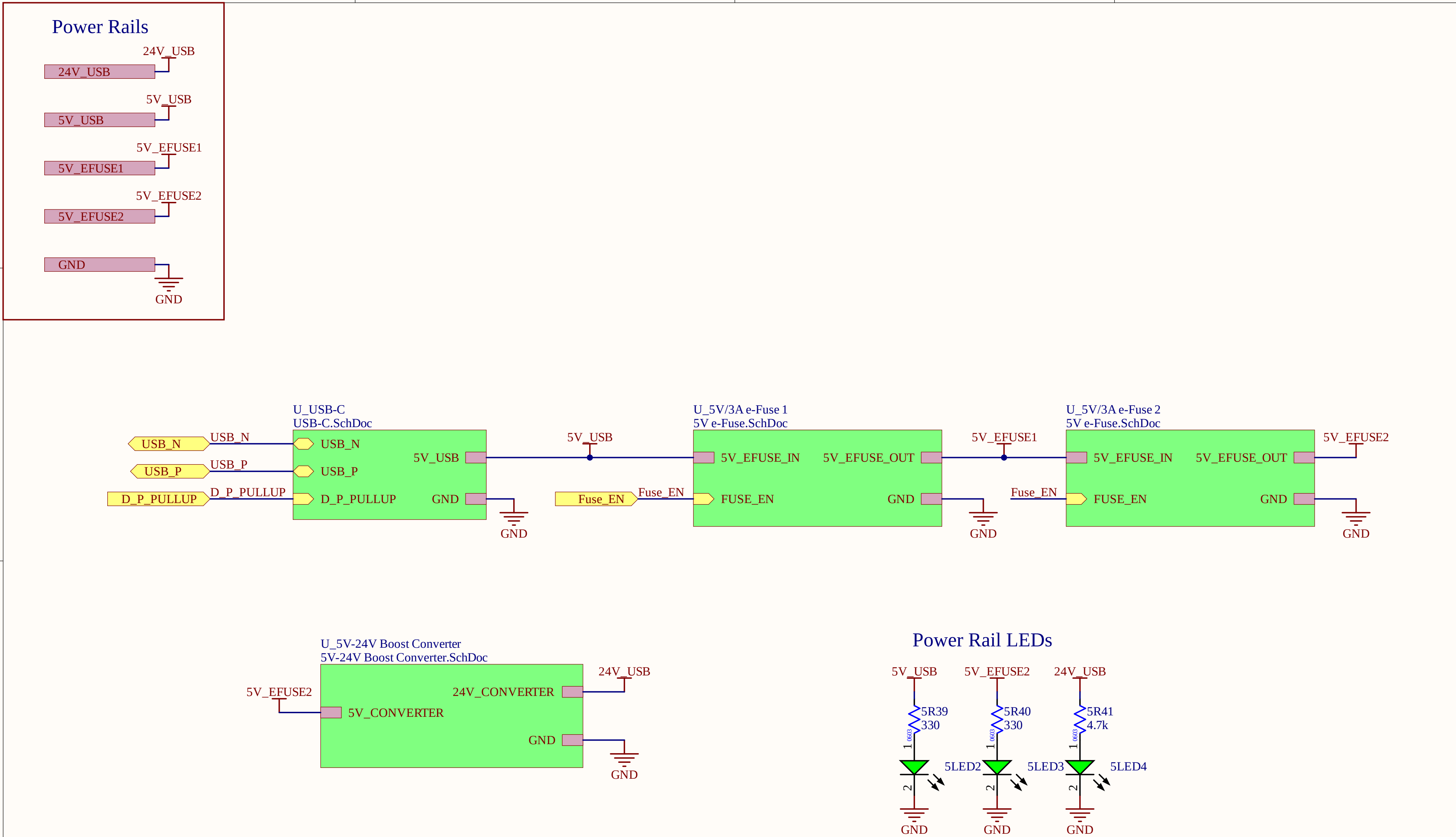
VCC_Logic and Fuse_EN signals can be configured by the board designer to control whether the debug board supplies power to the target board via USB-C.

On the debug board, Fuse_EN can enable or disable all e-fuses. VCC_Logic is connected to the 5V power rail of the USB-C.

These signals are shorted together in this schematic to signal for power by default.



UBC Formula Electric	Schematic Title: Off-Board HDMI (Debug Board Side)	Sheet #: 2 of 8
	Project Name:	Revision:
2345 E Mall Vancouver, BC, Canada electrical@ubcformulaelectric.com	Drawn By: Xinyi Zhao	Date: 6/14/2026



Schematic Title: USB-C Power and Signal	Sheet #: 3 of 8
Project Name:	Revision:
Drawn By: Xinyi Zhao	Date: 6/14/2026

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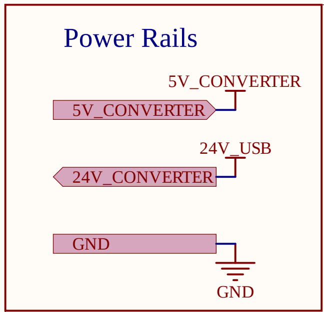
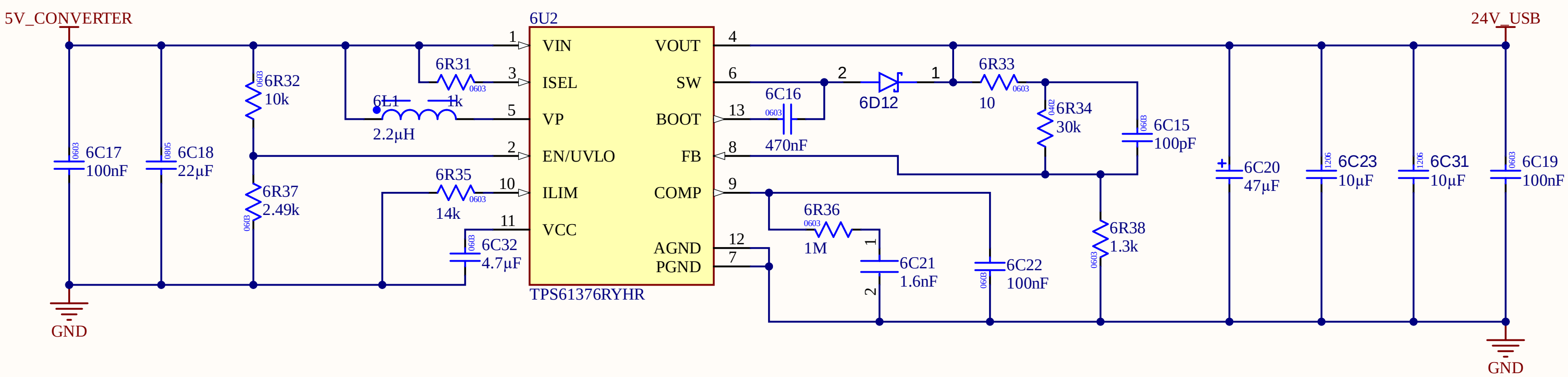


Vin = 5V
Vout ~ 24.307V
Vuvlo = 4V
ILIM (Input current limit) ~ 3.08A
Switching Frequency = 1200Hz
Cutoff Frequency = 30KHz



Schematic Title: 5V-24V Boost Converter	Sheet #: 6 of 8
Project Name:	Revision:
Drawn By: Xinyi Zhao	Date: 6/14/2026

UBC Formula Electric

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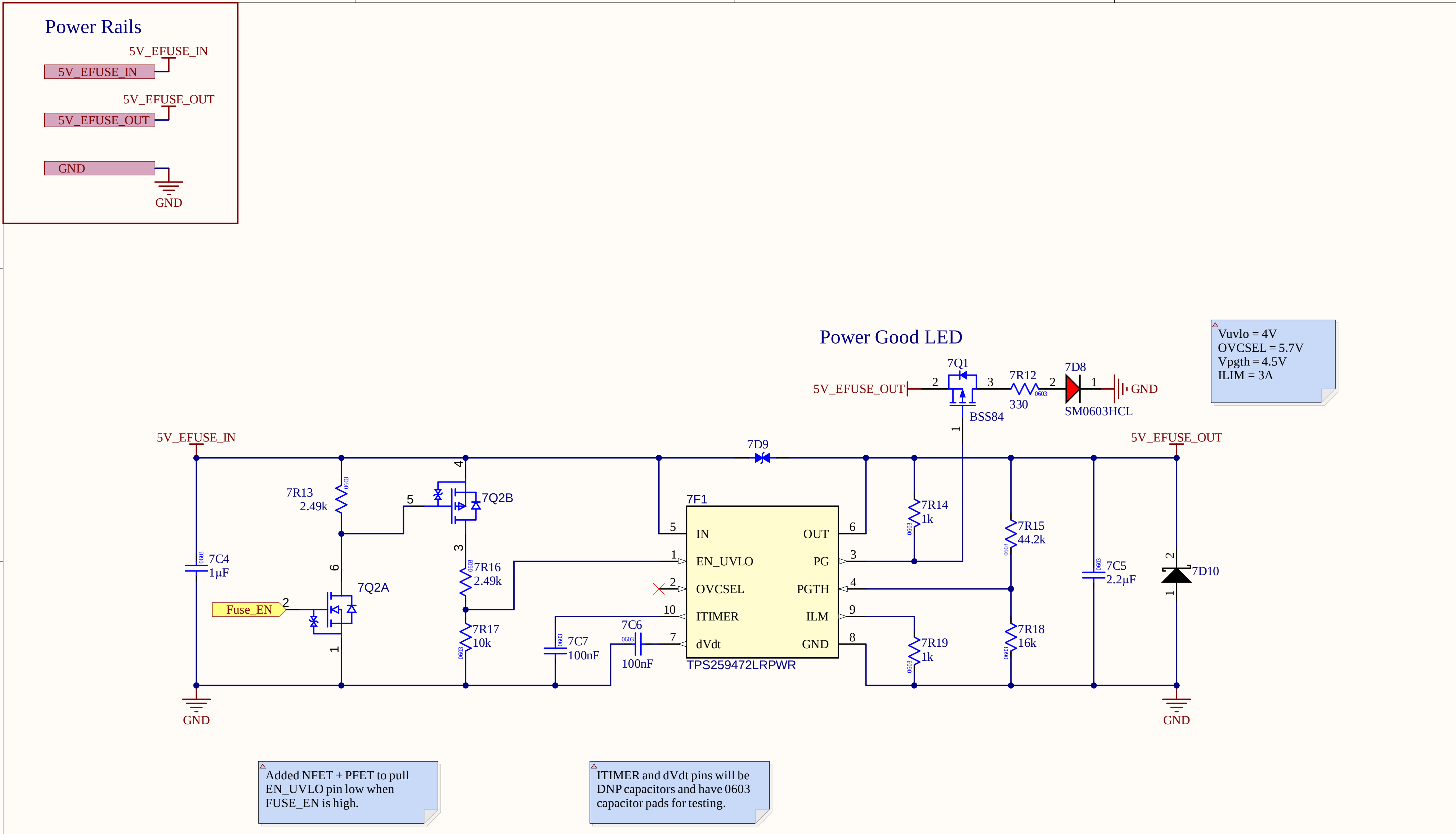
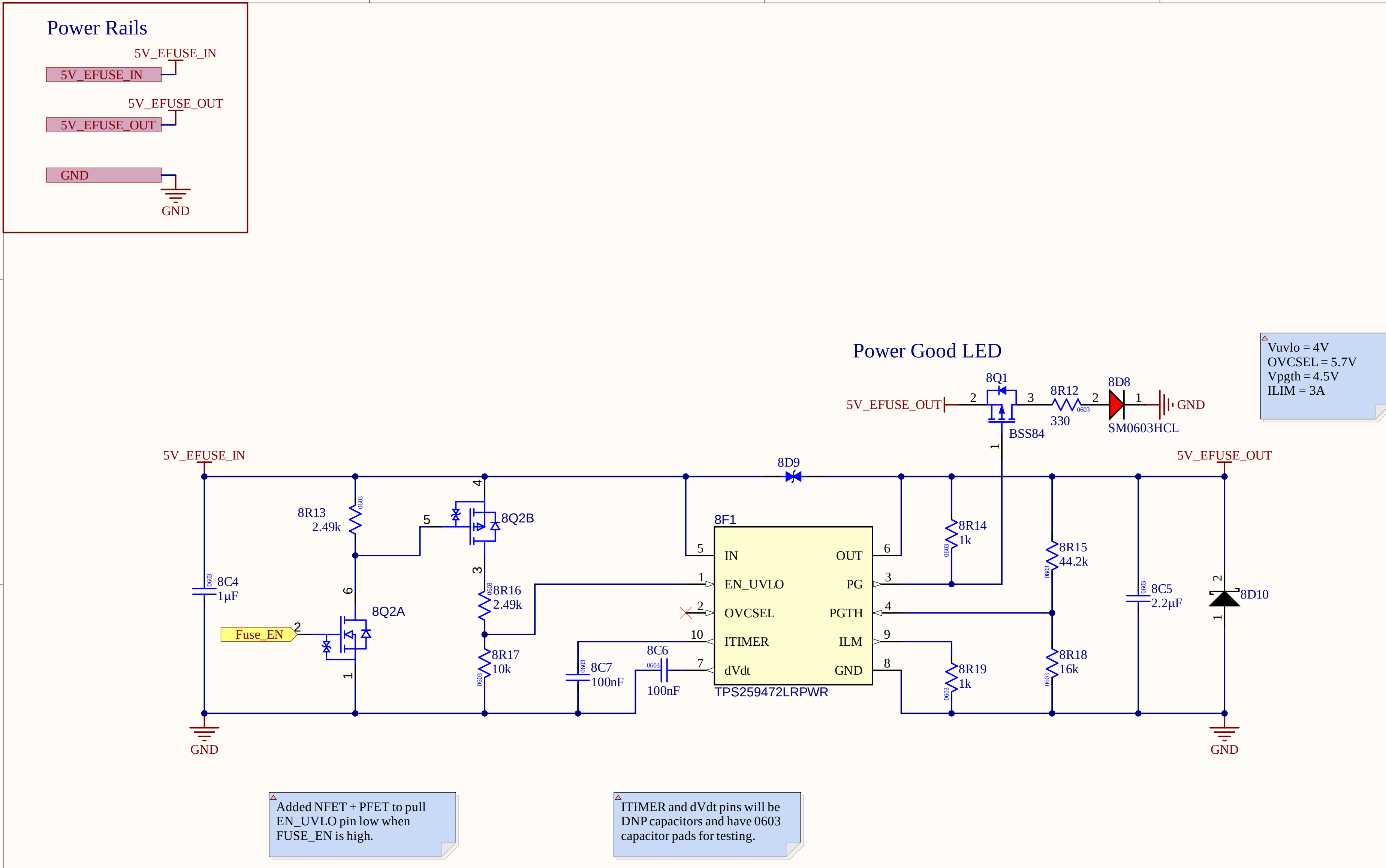
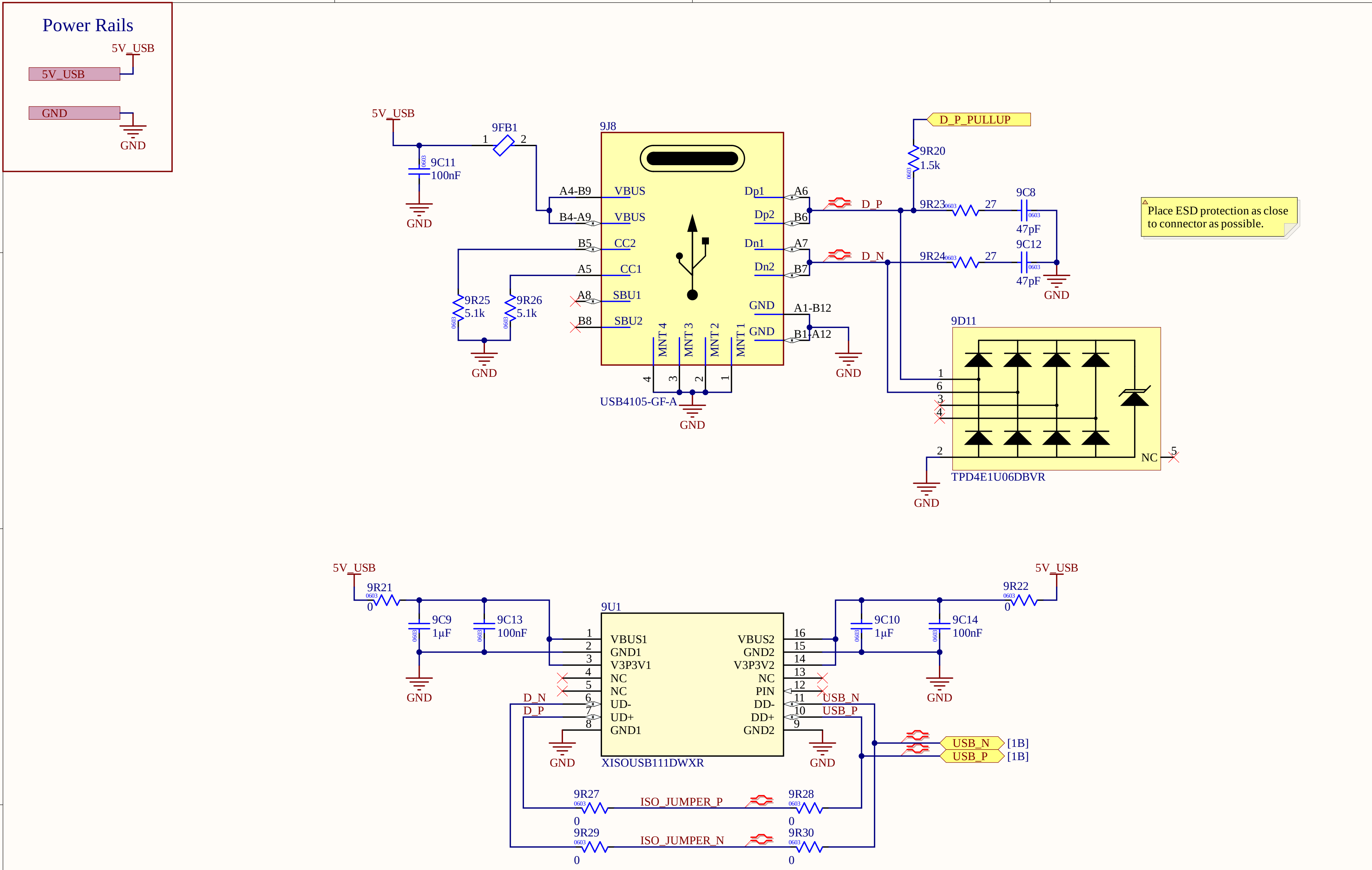
electrical@ubcformulaelectric.com



UBC Formula Electric	Schematic Title: USB 5V/3A E-Fuse	Sheet #: 4 of 8
	Project Name:	Revision:
2345 E Mall Vancouver, BC, Canada electrical@ubcformulaelectric.com	Drawn By: Xinyi Zhao	Date: 6/14/2026



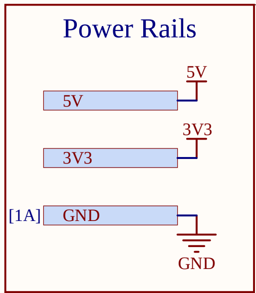
Schematic Title: USB 5V/3A E-Fuse		Sheet #: 4 of 8
Project Name:		Revision:
2345 E Mall Vancouver, BC, Canada electrical@ubcformulaelectric.com	Drawn By: Xinyi Zhao	Date: 6/14/2026



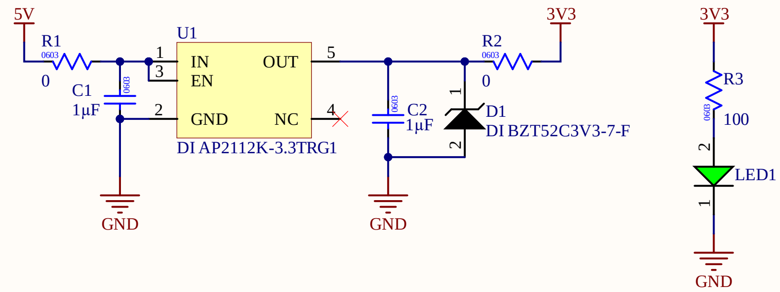
Place ESD protection as close to connector as possible.

R27, R28, R29, and R30 are DNP -- used to bypass U1 in case it malfunctions.

UBC Formula Electric	Schematic Title: USB-C Port	Sheet #: 5 of 8
	Project Name:	Revision:
2345 E Mall Vancouver, BC, Canada electrical@ubcformulaelectric.com	Drawn By: Xinyi Zhao	Date: 6/14/2026



5V to 3V3 Linear Regulator



Schematic Title: 5V to 3V3 Linear Regulator	
Project Name: Managed Schematic	
2345 E Mall Vancouver, BC, Canada electrical@ubcformulaelectric.com	Drawn By: Joe Thurston — Date: 6/14/2026